Pyuskulyan, Arthur
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https://arthurpyuskulyan.github.io/

Education

- **University of California, Merced** **Merced, CA**
  Ph.D. Chemistry Expected 2027
  University of California, Merced **Merced, CA**
  M.S. Chemistry 2021 – 2025
- **University of California, Irvine** **Irvine, CA**
  B.S. Physics 2019

Research Experience

- **Department of Chemistry, UC Merced** **Merced, CA**
  Graduate Student Researcher, Professor Christine Isborn's group 7/2021 – current
  -Developing and applying condensed phase molecular dynamics simulations, ground and excited state calculations, linear and nonlinear optical spectroscopy to investigate solvent effects on excited states and spectra
  -Writing Python scripts for simulation workflows, spectroscopy analysis, and large scale data processing
- **Department of Chemistry, and Physics and Astronomy, UC Irvine** **Irvine, CA**
  Undergraduate Researcher, Professor Kieron Burke's group 3/2018 – 7/2019
  -Performed density functional theory (DFT) calculations on diamagnetic molecules to compute electric field gradients, nuclear magnetic resonance (NMR) chemical shifts, and magnetic anisotropies using various functionals and pseudopotentials
  -Evaluated the accuracy of functionals and pseudopotentials through spreadsheet analysis, scripting, and statistical methods
- **Department of Physics and Astronomy, UC Irvine** **Irvine, CA**
  Undergraduate Researcher, Professor Jing Xia's group 3/2017 – 3/2018
  -Synthesized crystals and prepared low-temperature experiments in a dilution refrigerator to study topological insulator materials
  -Performed spot welding, soldering, experimental setup preparation, and data analysis for condensed matter physics experiments

Technical Skills

- Programming Languages: Python, Mathematica, LaTeX, Bash
- Technical Programs: TeraChem, Gaussian, AMBER, Quantum Espresso

Presentations

- "Static and Dynamic Condensed Phase Linear and Nonlinear Optical Spectroscopy Through Second-Order Cumulant-Based Correlation Function and Franck-Condon

Methods," poster at the West Coast Theoretical Chemistry Conference, UC Merced, California May 2024
- "Condensed Phase Optical Absorption and Emission Spectra: Comparison of Trajectory-based Correlation Function and Franck-Condon Methods," oral presentation at the American Chemical Society Spring Meeting, New Orleans, Louisiana March 2024
- Condensed Phase Optical Spectroscopy Modeled by Dynamic Cumulant Correlation Function and Static Ensemble Franck-Condon Methods," poster at the Time Dependent Density Functional Theory School and Workshop, Rutgers University, New Jersey June - July 2023
- "Predicting Nuclear Magnetic Resonance Parameters in Ceramics Using Density Functional Theory," poster at the SoCal Undergraduate Chemistry Research Symposium, UC Irvine, California 2019
- "Predicting Nuclear Magnetic Resonance Parameters in Ceramics Using Density Functional Theory," poster at the University of Southern California's SoCal Theo Chemistry Conference, University of Southern California May 2019
- "Band Gap Evolution in Ytterbium Hexaboride Under Uniaxial Strain," poster at the UC Irvine's Undergraduate Research Symposium 2018

Honors and Awards

- Chemistry and Biochemistry department travel award — 3/2024
- Chemistry and Biochemistry department summer fellowship — Summer 2023
- Chemistry and Biochemistry department travel award — 6/2023
- UC Irvine Undergraduate Research Opportunities Program research grant — 2018
- National Science Foundation S-STEM Program Scholarship — 9/2017 – 6/2019

Teaching and Service

- **American Chemical Society** — **Merced, CA**
  Mentor for Summer Experiences for the Economically Disadvantaged — Summer 2024
- **American Chemical Society** — **New Orleans, Louisiana**
  Spring Meeting session presider — March 2024
  -Presider for a Quantum Mechanics section
- **University of California, Merced** — **Merced, CA**
  Teaching assistant
  Chem 1 Preparatory chemistry — Fall 2022, Spring 2023
  Chem 2 Honors General Chemistry I — Fall 2024
  Chem 10 General Chemistry II — Fall 2021, Spring 2022, Spring 2026
  Chem 10H Honors General Chemistry II — Spring 2025
  Chem 112 Quantum Chemistry and Spectroscopy — Fall 2023
  Chem 130 Organic Spectroscopy and Computation — Spring 2023
- **University of California, Merced** — **Merced, CA**
  Secretary of the graduate chemistry club Ground State — Spring 2022 – Fall 2024
  -Helped organize professional and departmental events

- **Resonance Mentoring Program, Department of Physics, UC Irvine** **Irvine, CA**
  Mentor 1/2018 – 12/2019
  -Mentored and tutored freshmen to junior students
- **Grade Potential Tutoring** **Los Angeles, CA**
  Tutor 12/2019 – 4/2021
  -Tutored high school and college students with their physics, chemistry, and math courses

Publications

Christopher A. Myers, Shao-Yu Lu, Sapana Shedge, **Arthur Pyuskulyan**, Katherine Donahoe, Ajay Khanna, Liang Shi, Christine M. IsbornAxial H-bonding Solvent Controls Inhomogenous Spectral Broadening, Peripheral H-bonding Solvent Controls Vibronic Broadening: Cresyl Violet in Methanol, accepted May 2024 in *Journal of Physical Chemistry B*